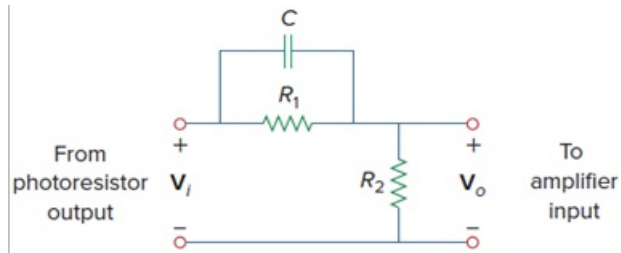


Problem

The RC circuit in Fig. 14.111 is used for a lead compensator in a system design. Obtain the transfer function of the circuit.

Figure 14.111:



Step-by-step solution

Step 1 of 1

$$H(\omega) = \frac{V_o}{V_i} = \frac{R_2}{R_2 + R_1 \parallel \frac{1}{SC}} \quad S = j\omega,$$

$$H(\omega) = \frac{R_2 \left(R_2 + \frac{1}{SC} \right)}{R_2 + R_1 \left(\frac{1}{SC} \right)}$$

$$H(\omega) = \frac{R_2 (1 + SCR_1)}{R_1 + SCR_2}$$